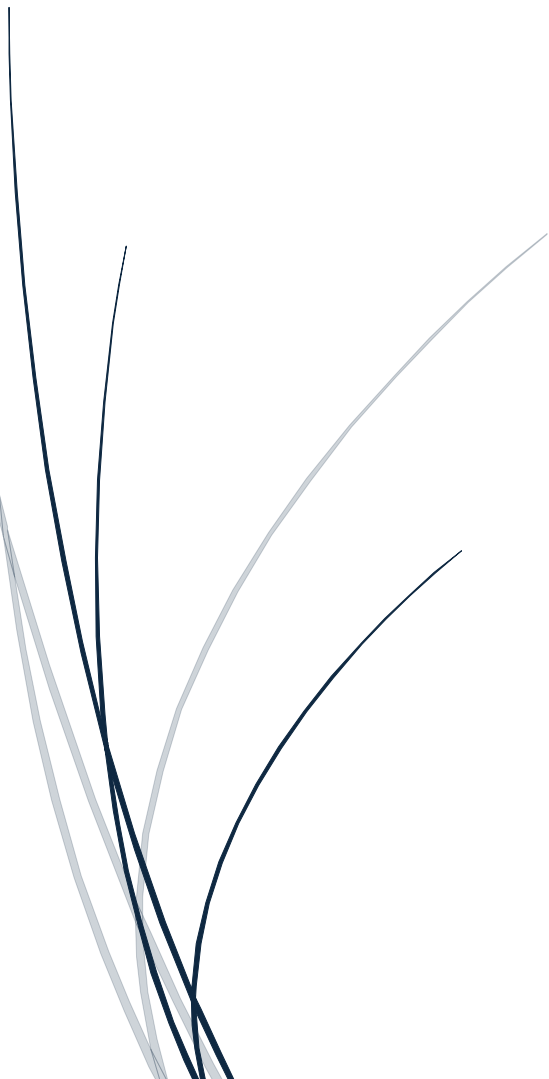


Research Proposal & PRD- designing

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Version Control

Version	Addition
0.1	First draft
0.2	Second draft

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1. AIBS – Research Proposal (first draft)

1.1 Target Audience

This research focuses on Dutch mechanical engineering and industrial product design firms with 20–75 employees that use CAD software such as SolidWorks, Autodesk Inventor, and Siemens NX. These companies want to determine whether AI can improve their design pipeline by reducing design time, improving design quality, and automating repetitive engineering tasks, while ensuring that the benefits outweigh the costs of implementation.

However, the growing integration of AI into 3D design software may increase dependence on a small number of mainly U.S.-based software providers, as many advanced AI features are developed by the software companies themselves. To reduce this dependency and strengthen strategic autonomy, organizations should explore alternative AI models, open-source solutions, and European or other non-American technology ecosystems. Diversifying AI providers can improve resilience, reduce supplier dependence, and increase long-term flexibility.

1.2 What do these companies need?

We see that companies are too afraid to start using AI in their business. Because they think the way of working now is perfect for them. There is also a lack of knowledge about the things AI can do, by the people who make these decisions in these businesses. But they are forced to change so they need to decide are we going to transform our company and be a more future realistic company, or fall behind and let other rival companies capitalize in their market.

According to Draghi the innovation gap between the US/China and Europe is big so companies in Europe need to decide if they want to close that gap or not. Because it's in their own hands if companies want to be more successful according to Draghi they need to increase their investment more in AI, digital technologies, research and innovation. Instead of focussing on the present and results.

1.3 Questions & Methodology

1.3.1 Main question

How can designers and engineers use AI to improve their design pipeline?

1.3.2 Sub questions

Sub-question 1: What does a design engineer do on a daily bases?	
Sub-question type	Field
Goal	Current work for a design engineer
Research-strategy	Document analysis - Interview

Sub-question 2: What AI tools do exist for a Designers and Engineer?	
Sub-question type	Desk & Field
Goal	What is already available and how dan this be used.
Research-strategy	Document analysis - Interview

Sub-question 3: What parts of the designing can be supported or automated by AI?	
Sub-question type	Desk & Field
Goal	Insight in the possibilities of AI to support or automate task for design-engineers
Research-strategy	Document analysis - Interview

Sub-question 4: What do SME decision-makers currently know about AI applications for design engineering, and where are the gaps?	
Sub-question type	Desk & Field
Goal	Insight in the possibilities of AI to support or automate task for design-engineers
Research-strategy	Document analysis – Interview/Questionnaire

Sub-question 5: Has any comparable mechanical-engineering or product-design SME already implemented AI in their design pipeline, and what were the results?	
Sub-question type	Desk & Field
Goal	Identify a real case where AI has already been used in a comparable design pipeline, to compare against perceived opportunities and risks.
Research-strategy	Document analysis (case studies, industry press) - Interview

Sub-question 6: What opportunities and risks brings AI for a designers and Engineers?	
Sub-question type	Desk & Field
Research-strategy	Document analysis - Interview
Goal	Get insight in how design-engineers see the opportunities and risks in the use of AI in their work

1.4 Quality criteria for the handbook

- **Understandable**
 - Engineers and designers must be able to understand the content without needing AI expertise.
 - Technical terms are explained when necessary.
- **Practically applicable**
 - The handbook provides recommendations and examples that can be directly applied in daily design work.
- **Balanced**
 - Both opportunities and risks of AI are discussed, allowing readers to make informed decisions.
- **Action-oriented**
 - Readers understand what actions they can take to start using AI in their work.
- **Up-to-date**
 - The handbook includes current AI tools, technologies, and developments.
- **Well-structured**
 - Information is logically organized and easy to navigate.
- **Easy to use**
 - Readers can quickly find the information they need through clear headings, page numbers, and a table of contents.
- **Evidence-based**
 - Recommendations are supported by research findings and interview results.
 - This is reinforced by Sub-question 5, which compares design-engineers' perceived opportunities and risks against a documented case of a comparable SME that has already implemented AI in its design pipeline.

1.5 Company interview

Companies that exist and we could interview

Name	Location	What do they do?	Why are they interesting for us and how are they using a design engineer?
CAD2M	Doetinchem	Works with SOLIDWORKS, CAD/CAM, PDM, 3D design and engineering software.	Very relevant because they work directly with CAD and are also interested in AI and digitalisation in engineering. Create and modify 3D CAD models, support customers with SOLIDWORKS, CAD/CAM and PDM software, improve digital design workflows, and advise companies on new technologies such as AI in engineering.
VIRO	Arnhem	Engineering company working on mechanical design, machine building and industrial projects.	Good example of how mechanical engineers use CAD and new digital tools in daily engineering work. They also work with students.

		Uses software such as SolidWorks, NX, Solid Edge and Inventor.	Design machines and industrial systems, create 3D CAD models and technical drawings, perform engineering calculations, and develop custom solutions for clients.
<u>Gusella Bakker B.V.</u>	Elst	Designs and manufactures hydraulic attachments and equipment for cranes and excavators.	Interesting because they combine mechanical design, CAD and actual manufacturing. We can ask how AI could improve their design and production process. Design hydraulic attachments for cranes and excavators, create production drawings, improve existing products, support manufacturing, and test new designs.
<u>MPS Systems B.V.</u>	Arnhem	Designs and manufactures complex printing presses for the global label industry.	Interesting because they combine mechanical engineering, automation, control systems and manufacturing. Design components and complete printing presses, develop mechanical systems, collaborate with automation and software engineers, and optimize machine performance and reliability.
<u>Kelvion</u>	Doetinchem	Designs and manufactures heat exchangers and industrial thermal equipment.	Their engineers use SolidWorks and 3D CAD, so they can give insight into how AI may support mechanical design and engineering calculations. Design heat exchangers and thermal systems, perform engineering calculations, create CAD models, select materials, and improve energy efficiency and product performance.
<u>Von Gahlen</u>	Zevenaar	Designs and builds specialised mechanical and mechatronic equipment.	Interesting because they combine mechanical, electrical and software engineering, which makes AI applications especially relevant. Design specialized mechanical and mechatronic equipment, integrate mechanical, electrical and software components, develop prototypes, and test new solutions.
<u>Royal SMIT Transformers</u>	Nijmegen	Designs and manufactures large power transformers and electrical equipment.	Interesting because they use advanced 3D CAD and have a large engineering department where AI could be used for design, calculations and optimisation. Design transformers and electrical equipment, create 3D models and technical drawings, perform engineering calculations, optimize designs, and support manufacturing processes.

1.6 Internal & External Analysis: DESTEP, 7P and SWOT

Supporting analysis for the research proposal and PRD of the AI handbook for Dutch mechanical engineering and industrial product design SMEs (20–75 employees). DESTEP covers the external environment, 7P covers the handbook and its research platform as a product, and the SWOT brings both together.

1.6.1 External analysis DESTEP

Six external factors shaping how quickly the target audience will engage with AI, and the environment the handbook is published into.

FACTOR	DESCRIPTION
Demographic	An ageing workforce and a shortage of skilled mechanical/CAD engineers. Experienced designers are retiring and taking tacit design knowledge with them, while the younger engineers replacing them are more open to AI-supported tools — creating both urgency and an opening.
Economic	CBS reports AI adoption among Dutch companies (10+ employees) rising from 14% in 2023 to 22.7% in 2024, but adoption scales with size: ~17.8% at 10–19 employees versus 59.2% at 500+. Our 20–75 target group is under-represented. A typical SME implementation costs €5,000–10,000, so cost is a modest barrier; unclear ROI is the bigger one.
Social / Cultural	Engineering culture here is risk-averse and quality-driven, and AI output touching technical drawings or safety-relevant components raises legitimate trust concerns. Decision-makers rarely know what AI can and cannot do in a design context, producing a wait-and-see attitude rather than active resistance.
Technological	SolidWorks, Autodesk Inventor and Siemens NX are embedding generative design and AI-assisted modelling directly into their platforms, so firms need no separate AI infrastructure. CBS data shows the largest barrier is experience, not cost: roughly 74–75% of non-adopters cite missing knowledge or skills.
Ecological	Generative design and AI-assisted optimisation can reduce material use and waste, aligning with EU sustainability and circular-economy pressure. This gives AI adoption a secondary justification for firms that must report on sustainability performance.
Political / Legal	The EU AI Act's high-risk obligations became enforceable on 2 August 2026, with penalties up to €35 million or 7% of global turnover, raising the stakes of adopting AI without understanding it. Meanwhile national and EU programmes (NL AI Coalition; the Digital Decade target of 90% of SMEs at basic digital intensity by 2030) are actively funding exactly this target group.

1.6.2 Internal analysis 7P

The marketing mix applied to the handbook and its supporting research platform as the product of this project.

P	ANALYSIS
Product	An English-language handbook, plus the research platform that produces it, explaining in practical terms how AI can support the design pipeline of mechanical engineering and product-design SMEs. Every claim is traceable to a source, so the product is curated and verified knowledge rather than plain information.
Price	No sales price — this is a research deliverable. The audience pays in time and attention; internally the cost is researcher time spent collecting, translating and verifying sources. The barrier to use stays low, but there is no revenue to sustain updates after the project ends.
Place	Distribution reuses existing wiki and website infrastructure (a project constraint), avoiding the cost of building a new channel. Reach therefore depends on how well that platform is already known to design engineers and SME owners.
Promotion	Awareness comes mainly from the research interviews: each interviewed company becomes a stakeholder with early exposure to the findings, supporting word-of-mouth via sector organisations and professional networks. There is no dedicated plan yet for SMEs outside the research sample.
People	A four-person student team collects and analyses sources and runs the platform. Primary users are non-technical researchers doing the gathering; secondary users are SME owners and design engineers. Success depends on the team's ability to interview credibly and judge source quality, since it is not yet domain-expert in mechanical engineering.
Process	A six-step workflow: collect sources, analyse, translate, generate a draft, human review, publish. Because translation and draft generation are both automated and error-prone, human review is the explicit quality gate before publication.
Physical evidence	The handbook pages themselves — consistent structure, APA references, HAN branding — and the published wiki presence. Consistency of tone, terminology and citation style signals rigor to a reader deciding whether to act on the content.

1.6.3 SWOT analysis

Strengths and weaknesses follow from the 7P analysis; opportunities and threats from DESTEP.

STRENGTHS (INTERNAL) • Academically grounded method: findings rest on document analysis and interviews, and every claim is traceable to a source. • Reuses existing wiki infrastructure, so no time goes into building a distribution channel. • Pre-defined quality criteria (understandable, practical, balanced, action-oriented, evidence-based) guide content from the start. • A four-person team lets research, translation, platform work and writing run in parallel.	WEAKNESSES (INTERNAL) • No budget or revenue model to maintain the handbook after the project ends. • The team is not yet domain-expert in mechanical engineering, risking missed nuance in technical sources. • Automated translation and AI-generated drafts carry a real risk of mistranslated terms and hallucinated claims. • Reliance on an agentic CLI rather than custom software limits control over workflow and output quality.
OPPORTUNITIES (EXTERNAL) • Dutch AI adoption is accelerating (14% → 22.7% in one year), creating a growing, receptive audience. • NL AI Coalition and EU Digital Decade programmes are funding exactly the SMEs we target. • AI is being built into the CAD platforms these firms already use, lowering the barrier the handbook must address. • The EU AI Act deadline creates urgency to understand AI responsibly — something the handbook can cover alongside opportunity.	THREATS (EXTERNAL) • The dominant barrier is missing skills (cited by ~three-quarters of non-adopters), which may also limit how much of the content lands without extra support. • Risk-averse engineering culture may cause readers to dismiss AI recommendations as impractical for safety-critical work. • AI capability evolves fast, so tool-specific references risk dating quickly. • Consultancies (€5,000–10,000 per SME project) and CAD vendors already compete for the same audience's attention.

Strategic implications

- Lead the handbook with verified, sourced examples rather than general claims, using the evidence-based method to counter a risk-averse audience.
- Turn the NL/EU funding programmes into a concrete recommendation section, since the main barrier is knowledge rather than budget.
- Treat human review as a formal QA gate, not an optional check, to offset the accuracy risk in translation and draft generation.
- Frame tool-specific content (SolidWorks, Inventor, NX) as examples of a broader trend, so the handbook dates more slowly.

2. AEL – Product requirements document

2.1 Problem and users

2.1.1 Problem

Researching AI applications for designers and engineers is time-consuming. To quantify this, we will first review AI features advertised by SolidWorks and other relevant engineering software vendors. We will then create a survey for HAN Mechanical Engineering and Automotive students, asking whether they use AI tools, how they use them, and how useful they believe these tools are. Where possible, the results of students who use AI will be compared with those who do not. This will help create a dataset showing which AI capabilities are being used and how effective they may be in engineering and CAD.

One limitation of this approach is that the test group is highly specialized, as all participants are students from similar technical study programs. This may introduce bias and means that the results may not represent professional engineers or the wider industry. This limitation could be mitigated by expanding the survey to other groups like employees of companies we interview or attempt to run the survey in social media.

The research team needs a platform that can efficiently collect, translate, analyse, and organize information into handbook pages.

2.1.2 Primary Users

- Student research team members
- Non-technical researchers who need to gather and analyse sources

2.1.3 Secondary Users

- SME owners
- Design engineers

These users will ultimately read the handbook to understand how AI can improve their design pipeline.

2.2 What must the user do

The platform should support the complete workflow from source collection to handbook publication.

1. Collect Sources	• Upload documents • Add website URLs • Add interview transcripts • Import Dutch and English sources
2. Analyse Sources	• Summarize documents • Extract key findings • Identify opportunities and risks • Detect information relevant to the research question

3. Translate Content	• Translate Dutch sources into English • Preserve technical terminology • Show original text alongside translations
4. Generate Handbook Draft	• Create handbook page drafts • Generate APA references • Suggest figures and examples
5. Human Review	• Edit generated prompt • Verify sources • Approve final handbook pages • Check wording against ISO 20607 4.4 comprehensibility principles
6. Publish	• Export handbook pages

2.3 Output Qualities

A handbook page produced by the platform should:

Output quality	Description
1. Be structured consistently	Each page follows a standard format, including: • Title • Introduction • Main findings • Practical examples • Recommendations • References
2. Be written in clear English	The content on the page is written in professional, easy-to-understand English suitable for engineers, designers, and SME owners. Terminology and wording follow the comprehensibility principles described in ISO 20607:2019, Clause 4.4 (Comprehensible terminology and wording).
3. Be traceable to sources	Every key claim, recommendation, and finding is linked to a supporting source or interview.
4. Maintain a consistent style	Pages use the same tone, terminology, formatting, and citation style.
5. include citations	Sources are referenced according to APA 7th edition.

2.4 Known constraints

Requirements for literature / sources

- Literature is no more than 4 years old.
- Sources are academic or professional (no Wikipedia or blogs without an author).
- Language: Dutch and/or English.
- Sources originate from recognized publishers, professional journals, government bodies, or reputable knowledge organizations.

Agentic CLI

- The platform must be operated through an agentic CLI instead of custom-coded software.

Existing Infrastructure

- The project must reuse existing wiki and website infrastructure.

2.5 Source material

Many useful sources for Dutch SMEs are written in Dutch, including:

- CBS reports
- Chamber of Commerce publications
- Sector organizations
- Regional news articles
- Interview transcripts

2.6 Out of scope (for now)

The platform will NOT:

- Build custom AI models
- Replace human decision making
- Conduct interviews itself
- Generate engineering designs
- Integrate directly into CAD software

Reason for excluding custom AI models: existing generative AI tools are already sufficient for the summarisation, translation, and extraction tasks this platform requires, and training custom models would require data, budget, and time disproportionate to a project of this scope. Off-the-shelf tools are therefore used as-is; this may be reconsidered in future versions.

2.7 Open questions and assumptions

2.7.1 Open questions

- Has any comparable mechanical-engineering/product-design SME already implemented AI in its design pipeline? How will such a case, if it exists, be found and verified?

- What counts as a "successful" reduction in research time is there a target percentage or number, or just "faster than the 37-minute baseline"?
- Who maintains or updates the handbook after the project ends, given there's currently no budget or owner for that (flagged as a weakness in the 7P/SWOT work)?

2.7.2 Assumptions

- Decision-makers at target SMEs currently lack sufficient knowledge of AI's capabilities not yet tested, currently just asserted in 1.2.
- The innovation gap Draghi describes between the EU and US/China applies meaningfully at the level of individual CAD-tool AI features, not just macro R&D investment.
- SME decision-makers will actually read and act on a written handbook, rather than needing hands-on consultancy support to change behaviour.

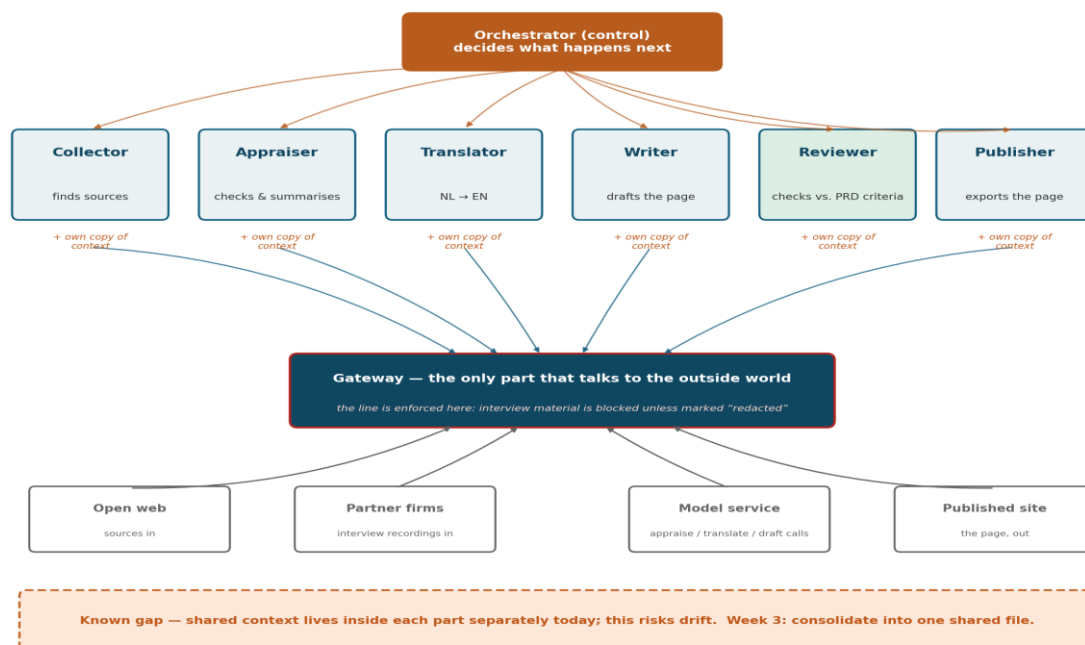
3 Technical Blueprint

This blueprint implements the AEL PRD. It draws the platform at two altitudes, records the team's six decisions about its shape, names the file contract at one seam, the one part that enforces the line to the outside world, and where every PRD promise lands.

3.1. Two altitudes

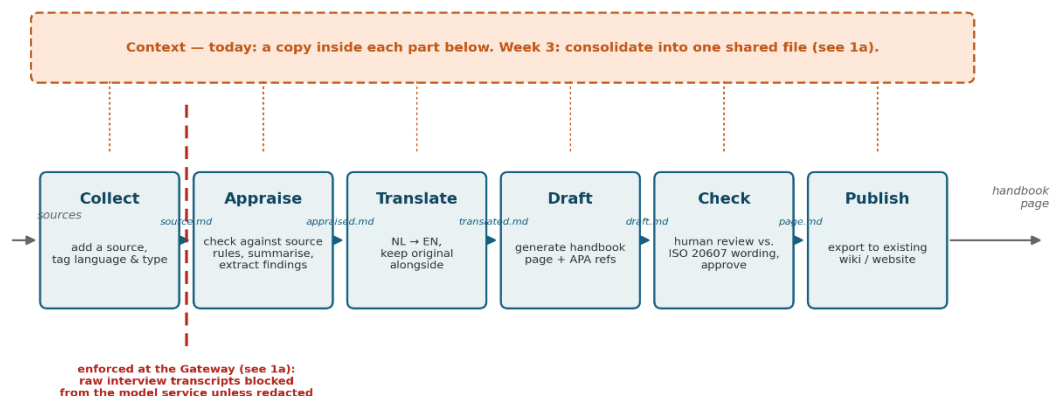
3.1.1 The company who does what

Roles, control, and the current state of shared context including the known gap named in Decision 4. Every external call funnel through one part: the Gateway.



3.1.2 The production line what follows what

The same six roles, now ordered, with the file each stagehands to the next. This altitude answers what has to happen before what.



3.2. The six decisions

Answered together as a team, in our own words.

Decision 1: Parts

We decided: Our parts are roles: each one has a job we can write in a single sentence. Collect, Appraise, Translate, Draft, Check, Publish.

Decision 2: Control

We decided: One part decides what happens next and calls the others in order.

Decision 3: Contracts

We decided: Parts hand over a file with a fixed, named shape, which is the contract between them: source.md → appraised.md → translated.md → draft.md → page.md.

Decision 4: Shared context

We decided: Shared context currently lives inside each part separately, which we know risks drift, and week 3 will fix it.

Decision 5: The line

We decided: One part is the only one that talks to the outside world, and the line is enforced there. The Gateway is the sole point of contact with the open web, partner firms, the model service, and the published site; interview material is blocked from reaching the model service unless it has been marked redacted.

Decision 6: Being wrong

We decided: A checking part holds output against the criteria in our PRD before it moves on. Check compares each draft page against the 1.4 quality criteria and the ISO 20607 wording principles before Publish is allowed to run.

3.3 What passes between the parts: appraised.md

The file Appraise hands to Translate. Its shape is fixed so Appraise and Translate can be built by different people against the same agreement.

Field	Type	Meaning
source_id	string	Id assigned at Collect; every later file refers back to it.
source_type	enum	academic professional government vendor interview — checked against the 2.4 sourcing rule.
language	enum	NL EN— read by Translate to decide whether translation is needed.
publication_date	date	Used to enforce the “no more than 4 years old” rule (2.4).
summary	string	Written by Appraise.
key_findings / opportunities / risks	list<string>	Written by Appraise; each item traceable back to source_id, feeding AIBS sub-questions 4 and 6.

review_status	enum	pending redacted approved rejected — interview sources must reach “redacted” before Appraise may pass them onward.
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3.4 The line nothing may cross

Interview material never reaches a model service. The recordings are allowed inside the platform they are the team's own material, gathered from the companies named in section 1.5 but only the Gateway may pass anything to a model service, and the Gateway blocks any source of type "interview" whose review status is not yet "redacted". A human does that redaction removing company and person identifying detail before the source can move past the Gateway. Because this is enforced in exactly one part rather than wherever a part happens to make an external call, it only needs to be built, and checked, once.

3.5 Where the map stops

Three things the ten classic architecture styles do not name. Decision 6 (Being wrong) already answers the first of these for us; the other two are not yet team decisions, so they're named here as open considerations for weeks 4-6.

Evaluation of who checks the work (Decision 6, restated)

Two checks, not one. Appraise runs an automatic check against the 2.4 sourcing rules (age, publisher type) before a source is used at all. Check is the human check against the 1.4 quality criteria and the ISO 20607 wording principles, and nothing reaches Publish without it.

Monitoring what happens when it is wrong (open)

Not yet a team decision. Our working assumption is that the default should be stopped, not publish: a source that fails the Appraise check returns to Collect with a reason attached rather than being silently dropped, and a page rejected at Check goes back to Draft with the reviewer's notes. This needs to be confirmed as a decision, not assumed.

Skills and memory what is carried, what is looked up (open)

Once shared context is consolidated (Decision 4, week 3), it becomes the thing every part carries by default. Everything else, a specific source's content stays looked up per source, never assumed to apply to another. The one thing we intend never to carry forward is raw, unredacted interview text.

3.6 The trace to the PRD

Every promise in the PRD lands in a part below, or the table says why not.

PRD promise	Lands in	Note
2.2.1–2.2.6 Collect → Publish workflow	Collect / Appraise / Translate / Draft / Check / Publish	Direct, one part per numbered step.
2.4 Sourcing rules, agentic CLI, existing infrastructure	Shared context + all parts / Publish	Rules belong on shared context (today: duplicated, see Decision 4); the CLI is how every part runs, not a part itself.
1.4 / 2.3 Quality criteria incl. ISO 20607	Check	The approval gate, not a separate step.
2.1.1 Timing benchmark	outside the pipeline	A one-off evaluation activity run around the platform, not a part.
2.6 HAN student survey	outside the pipeline	Separate research feeding shared context's assumptions, not a part.
1.1 Reducing dependence on US-based AI providers	not yet placed	Open: which model service the Gateway calls on behalf of Appraise/Translate/Draft. See PRD action below.
2.7 Out of scope items	deliberately absent	Confirmed as non-parts; already excluded by the PRD.

Departure and PRD update

Departure: the PRD (1.1) raises a concern about dependence on US-based AI providers, but this blueprint does not yet say which model service the Gateway calls on the platform's behalf.

Named here as an open decision rather than settled silently.

PRD action: revise the PRD to v0.3, dated after the team agrees this blueprint, adding one line to 2.4 naming the permitted model service(s) behind the Gateway and why visible as a constraint, not buried as an implementation detail.